

**CENG381 Stochastic Processes
Midterm Exam**

Duration: 90 minutes

Q1 (30 points). Three out of every four trucks on the road are followed by a car, while only one out of every five cars is followed by a truck. What fraction of vehicles on the road are trucks?

Q2 (35 points). A string transducer is defined over the alphabet $\{a, b\}$. It operates on input w by producing a string of length w whose i th symbol is an a if $i = 1$, or if $i > 1$ and the i th and $(i - 1)$ st symbols of w are different; otherwise, the i th symbol of the output is a b .

For example, on input $aabba$ the transducer should output $ababa$, and on input $aaaab$ it should output $abbba$.

If $i > 1$, producing a different letter with a state change has a probability $p = 0.3$, while producing the same letter with a state change has a probability $q = 0.4$.

Sketch the corresponding Markov chain for this transducer and find the limiting state probabilities.

Q3 (35 points). The Fibonacci numbers are 1, 1, 2, 3, 5, 8, 13, 21, 34, ..., where each number in the sequence is the sum of the two previous numbers. Show that the ratio of consecutive Fibonacci numbers tends to the "golden ratio" $\phi = (1 + \sqrt{5})/2$.